Unit: 6 Unit 2: Exponents Radicals
Topic: Exponents

Name: ________________
Date: ___________

Advanced (Carolina Reaper)

Q1. Simplify $2^3 \cdot 2^4$
(A) 2^5
(B) 2^6
(C) 2^7
(D) 2^{12}
(E) 2^{10}

Q2. What is the value of x in $x^2 = 64$?
(A) 8
(B) -8
(C) 4
(D) -4
(E) Both A and B

Q3. Simplify $(3^2)^3$
(A) 3^5
(B) 3^6
(C) 3^8
(D) $3^{2\cdot 3}$
(E) $3^{3\cdot 2}$

Q4. Simplify $\frac{10^3}{10^2}$
(A) 10^1
(B) 10^5
(C) 10^{3-2}
(D) 10^{2-3}
(E) 10^{3+2}

Q5. Tom has been saving money for a drone and has 2^5 dollars in his piggy bank. If he needs 2^6 dollars to buy it, how much more money does he need?
(A) 2^5
(B) 2^6
(C) 2^{6-5}
(D) 2^{5-6}
(E) 2^{5+6}

Q6. Sarah is a hiker and wants to calculate the area of a square trail. If each side of the trail is 5^2 meters, what is the area?
(A) 5^2
(B) 5^3
(C) 5^4
(D) $5^{2\cdot 2}$
(E) 5^{2+2}

Q7. What is the simplified form of $\frac{6^3}{6^2}$?
(A) 6^{3-2}
(B) 6^{2-3}
(C) $6^{3\cdot 2}$
(D) $6^{2\cdot 3}$
(E) 6^{3+2}

Q8. A water tank can hold 2^5 gallons of water. If 2^3 gallons are already in the tank, what fraction of the tank is filled?
(A) $\frac{2^3}{2^5}$
(B) $\frac{2^5}{2^3}$
(C) $\frac{2^{5-3}}{2^5}$
(D) $\frac{2^{3-5}}{2^3}$
(E) $\frac{2^{3+5}}{2^5}$

Open Ended Questions (FRQ)

Q9. Simplify the expression $\frac{(2^2)^3}{(2^3)^2}$ and show your work.

Q10. A bakery is having a sale on bread. A regular loaf normally costs 2^3 dollars, but it's on sale for 2^2 dollars. How much money will you save if you buy 2 loaves of bread during the sale? Show your work and calculate the total savings.

Chef's Tips

- Tip 1: Exponent rules can be applied in various real-life scenarios, such as finance and science.
- Tip 2: Simplifying expressions with exponents requires careful attention to the rules of exponents.
- Tip 3: Students should be encouraged to use examples from their daily lives to illustrate the practical applications of exponent rules.